

Multiple Choice (10 marks)

1. Which of the following is not a security exploit?
 - (a) Eavesdropping
 - (b) Cross-site scripting
 - (c) Authentication
 - (d) SQL Injection
2. _____ is the central node of 802.11 wireless operations.
 - (a) WPA
 - (b) Access Point
 - (c) WAP
 - (d) Access Port
3. You are given a message (m) and its OTP encryption (c). Can you compute the OTP key from m and c ?
 - (a) No, I cannot compute the key.
 - (b) Yes, the key is $k = m \text{ xor } c$.
 - (c) I can only compute half the bits of the key.
 - (d) Yes, the key is $k = m \text{ xor } m$.
4. What was the first buffer overflow attack?
 - (a) Love Bug
 - (b) SQL Slammer
 - (c) Morris Worm
 - (d) Code Red
5. Failed sessions allow brute-force attacks on access credentials. This type of attacks are done in which layer of the OSI model?
 - (a) Physical layer
 - (b) Data-link Layer
 - (c) Session layer
 - (d) Presentation layer

True / False (10 marks)

Answer True or False

6. True or False: The digest created by a hash function is commonly referred to as a Modification Detection Code (MDC).
7. True or False: The .Net framework is primarily designed for developing secure applications and does not focus on vulnerability exploitation.
8. True or False: EXE restarts the STP solver and continues executing the path, assuming the query is satisfiable despite the timeout.
9. True or False: WPA 2 is less secure than WEP and should not be used for protecting wireless networks.
10. True or False: Encapsulating Security Payload (ESP) is a component of the Secure IP Protocol, providing confidentiality and integrity for IP packets.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process and significance of mutation-based fuzzing in cybersecurity, focusing on how it generates inputs by modifying previous ones and its implications for software testing.
12. Discuss the classic security properties in cybersecurity, identifying which is not traditionally considered a core property, and analyze the implications of this distinction on security frameworks.

End of Paper